

Spring Boot 3 + JPA + YAML + Gradle - Summary & Flow

1. Entity Modeling

User entity with: id, name, email, deletedAt (soft delete), userRoles (one-to-many). UserRole entity with composite key via @EmbeddedId. Concepts: Entity mapping, soft delete, @OneToMany, @EmbeddedId.

2. DTO & Projections

DTO (constructor projection): UserDTO(id, name, email). Interface Projection: aggregates role count directly in query. Concepts: Spring Data projections, constructor vs interface projection.

3. Repositories & Queries

UserDTORepository: JPQL returning DTOs. UserSummaryRepository: JPQL + group by. UserJoinFetchRepository: entity queries with join fetch. Concepts: JPQL, pagination, fetch joins, aggregation.

4. Specifications

Reusable filters: nameContains, emailDomain, hasRole (join), notDeleted. Combined dynamically in controllers. Concepts: Criteria API, dynamic query building, reusable predicates.

5. RowMapper & JDBC

UserRowMapper for raw SQL → maps rows to User. Concepts: Spring JDBC interop, RowMapper pattern.

6. Database Schema

users table with deleted_at column. user_roles table with composite PK. Concepts: Soft delete at DB level, normalized relational schema.

Visual Flow of JPA + Spring Data

